SWAPNANIL SAHA

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PROFESSIONAL SUMMARY

I'm an M.Sc. Geoinformatics student with hands-on experience in flood and multi-hazard GIS mapping, DEM/LiDAR terrain analysis, raster and vector processing and Python-based geoprocessing automation. I've built spatial data pipelines with QA/QC and accuracy assessment (82-92% overall accuracy across hazard and LULC models) and applied AI/ML to geospatial hazard modelling. I use AI-assisted tools to accelerate Python scripting and testing and I'm eager to support flood hazard and catastrophe risk modelling.

SKILLS & EXPERTISE

GIS Platforms: ArcGIS Pro, ArcGIS Online, ArcGIS Experience Builder, ArcGIS Dashboard, QGIS, Google Earth Engine
Hazard & Terrain Analysis: Flood & Multi-Hazard Susceptibility Mapping, AHP-MCDA, DEM/DSM/DTM & LiDAR Processing, Terrain Indices (TRI, TWI, Drainage Density, Slope, Aspect)
Raster & Vector Analysis: Raster & Vector Data Processing, Spatial Analysis, LULC Mapping & Change Detection, Kernel Density, Focal Statistics, Spatial Joins
Programming & Automation: ArcPy & Open-Source Python (GeoPandas, Pandas, NumPy, Scikit-learn), GeoAI/ML (Random Forest, XGBoost), SQL (PostgreSQL, PostGIS - Basics), Geoprocessing Scripting, AI-Assisted Development & Testing
Data Management & QA/QC: GIS Data QA/QC, Accuracy Assessment, Documentation, Geotechnical Interpretation, Coordinate Reference Systems, GeoJSON, Shapefile, KML, CSV
Soft Skills: Analytical, Detail-Oriented, Collaborative, Fast Learner, Strong Written & Oral English Communication

EDUCATION

M.Sc. Geoinformatics, Symbiosis Institute of Geoinformatics (SIG), SIU | Pune 2025-2027
SGPA: 9.06 (Semester 1) & 9.15 (Semester 2) | Rank 2 in cohort
B.Sc. (Honours) Geography, Vidyasagar Metropolitan College, University of Calcutta | Kolkata 2022-2025
CGPA: 7.44 | Rank 2 in cohort

INTERNSHIP EXPERIENCE

Geospatial Project Intern | Department of Civil Engineering | Jadavpur University June 2026 - July 2026
- Developed a Python-based GIS and Machine Learning pipeline for Bengaluru urban water stress covering spatial data processing, feature extraction, variable forecasting and model development; published a Web GIS dashboard for stakeholders.

FEATURED PROJECTS

Predictive Urban Water Stress Mapping & Decentralised Infrastructure Siting in Bengaluru: A GeoAI Approach
Internship Project | ArcGIS Pro | Python | Prophet | ARIMA | Machine Learning | K-Means Clustering | Leaflet.js | TypeScript
- Engineered a Python pipeline integrating 11 geospatial variables (Rainfall, Temperature, LULC, Groundwater, Borewells, Pumping Stations, Water Pipelines etc.) to produce Random Forest and XGBoost Water Stress Maps for 2025 and 2035.
- Forecasted inputs: Rainfall (Prophet), LST (ARIMA), LULC (CA-Markov, 92.23% OA), Groundwater Depth (OLS); Allocated Recharge Wells, Storage Tanks and Rainwater Harvesting sites via K-Means (K = 3, Elbow, Silhouette & Davies-Bouldin).
- XGBoost outperformed with 87.97% Overall Accuracy & 0.819 Kappa; 35% of Bengaluru projected under severe water stress by 2035. **Dashboard Link:** https://water-stress-dashboard.vercel.app/

GLOF Risk Assessment using Cryospheric & Topographic Indicators with Random Forest: Kargil, Ladakh
Academic Mini Project | ArcGIS Pro | Random Forest | HTML & CSS
- Integrated 11 geo-environmental parameters (DEM, Slope, Aspect, TRI, TWI, MNDWI, NDSI, SWI, LULC, Glacier Extent, Glacial Lakes) with Random Forest for a GLOF Hazard Risk Map (82% OA, 0.680 Kappa).
- Quantified 24.6% Glacier Retreat (1049.94 km² to 791.72 km², 2000-2024) & mapped 579 Glacial Lakes: Periglacial (43.7%), Proglacial (36.8%), Supraglacial (19.5%). **Dashboard Link:** https://swapnanilsaha-gis.github.io/GLOF__Dashboard/

GIS-Based Asset Management Dashboard for a Natural Gas Pipeline Utility: Central Houston, Texas
Academic Mini Project | ArcGIS Pro | ArcGIS Online | ArcGIS Dashboards | Hosted Feature Services | Risk Modelling
- Published a 12-layer gas asset network (6,214 pipeline segments, 967 km, 3 pressure levels; city gates, storage, metering & pressure-regulating stations, valves, IoT sensors, consumers); a Risk = PoF × CoF model (9 weighted factors) flagged 946 High/Critical segments, 1,006 (16.2%) overdue inspections & $16.3M 5-year maintenance cost, ranking Top-10 priority segments.
- Designed a 3-view ArcGIS Dashboard with 6 KPIs, filters for 5 zones, 4 materials & 3 network levels, sensor alerts (pressure drop ≥ 20%/40%, CH_4 ≥ 2,500/5,000 ppm) driving 41 open maintenance requests, and cause analysis of 622 leaks.
- **Dashboard Link:** https://www.arcgis.com/apps/dashboards/ae23a5309f5f4d17b23d41f372590773

Mapping Emergency Service Deserts: A Spatial Intelligence Approach to Emergency Accessibility in Hyderabad, India
Academic Mini Project | ArcGIS Pro | ArcGIS Online | Experience Builder | Network Analysis | K-Means | GMM
- Built a 1 km hexagonal Accessibility Model across GHMC combining road-network distance to 4 emergency facility types (Hospitals, Fire, Police, PHCs), 2 vulnerability indicators (Slums, Flood Points) and 3 transit modes (Bus, Metro, Railway) using OSM networks and the Hyderabad Metro GTFS feed.
- Applied K-Means (K = 4, validated by Elbow & Silhouette) to classify 4 accessibility tiers, identifying 15% of GHMC as high-priority "Emergency Service Deserts"; designed and published a 12-page ArcGIS Experience Builder application.
- **Dashboard Link:** https://swapnanilsaha-gis.github.io/hyderabad-emergency-services-dashboard/

CERTIFICATIONS

Introduction to AI in ArcGIS | ESRI | Certificate January 2026
Python for Everyone | ESRI | Certificate November 2025
Basics of JavaScript | ESRI | Certificate October 2025
Introduction to LiDAR in ArcGIS Pro | ESRI | Certificate January 2026
Coordinate Reference Systems, Projections & Spatial Data Representation | Geo University | Certificate January 2026

ACHIEVEMENTS

- Finalist | Top 16 of 233 Teams | SPARK 2.0 | IEEE GRSS | Certificate
- 3rd Position | MAP MANIA 2025-26 | IIARI Kolkata | Certificate
- Rank 2 | M.Sc. Geoinformatics | Semester 1 & 2 | PG Merit Scholarship Receiver | SIG, Pune